

C3-092 Protected Review Pack

Open only after the matching Pxx attempt is saved

BOOK	->	QUALITY	->	OPERATE	->	PROVE
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2026 LEARNER EDITION

Option B institute family: Charcoal #1F2937 | Teal #14B8A6 | Soft Blue #3B82F6 | AMOLED-ready companion shell.

Protection rule

This is diagnostic feedback, not a solution bank. Compare only the weak competency, close the review, then solve the changed retry from a blank file/tab. Opening a future review early is a critical assessment-process failure.

R01 - Data-quality dimensions and enforceable contracts

Open only after genuine P01 evidence exists.

A strong attempt should show

A contract that names grain, stable key, required fields, allowed status values, owners, and at least four checks with explicit consequence.

Diagnostic questions

1. Did you declare the correct boundary/grain? 2. Does the evidence actually prove the claim? 3. Could a green job still hide the failure? 4. Is owner/action/recovery explicit? 5. Did you keep secrets/PII out of evidence?

Common miss

Writing checks before declaring grain; treating every NULL the same; using vague words like clean/good; having no owner when a rule fails.

Targeted repair

If you cannot classify a rule, restate the consumer impact and table grain before choosing severity.

Changed retry

Change the data to an event stream where one order legitimately has many status events: redesign the key and checks.

Close this review now. Save retry under evidence/L01/retry_001/ and explain what changed.

R02 - Blocking vs warning checks and severity design

Open only after genuine P02 evidence exists.

A strong attempt should show

A severity policy that maps each check to consumer impact, publish action, owner, escalation and recovery.

Diagnostic questions

1. Did you declare the correct boundary/grain?
2. Does the evidence actually prove the claim?
3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit?
5. Did you keep secrets/PII out of evidence?

Common miss

Making everything blocking; making everything warning; alerting without an owner; confusing test failure with pipeline failure.

Targeted repair

If everything blocks, distinguish business-critical invariants from triage-worthy anomalies. If nothing blocks, identify the corruption you are allowing through.

Changed retry

Reclassify one check after the consumer changes from exploratory dashboard to financial close.

Close this review now. Save retry under evidence/L02/retry_001/ and explain what changed.

R03 - Source-to-target reconciliation patterns

Open only after genuine P03 evidence exists.

A strong attempt should show

A report showing source/target counts, missing/extra keys, value mismatches and control-total delta at the stated grain.

Diagnostic questions

1. Did you declare the correct boundary/grain?
2. Does the evidence actually prove the claim?
3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit?
5. Did you keep secrets/PII out of evidence?

Common miss

Checking only row count; checking only total; using DISTINCT to hide duplicate keys; comparing different grains.

Targeted repair

If totals differ but you cannot explain why, compare key sets first and then row-level values for common keys.

Changed retry

Add one extra target row that offsets the missing-row total; show why key-set reconciliation still catches the defect.

Close this review now. Save retry under evidence/L03/retry_001/ and explain what changed.

R04 - Freshness, completeness and timeliness controls

Open only after genuine P04 evidence exists.

A strong attempt should show

Fixed-time freshness calculation, expected-partition completeness check, breach decision, owner and alert action.

Diagnostic questions

1. Did you declare the correct boundary/grain?
2. Does the evidence actually prove the claim?
3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit?
5. Did you keep secrets/PII out of evidence?

Common miss

Using current clock during tests; measuring job end instead of data time; ignoring expected missing partitions; paging on harmless delay.

Targeted repair

If freshness changes each time you run the test, you forgot to make observation time explicit. If completeness is just row count, define expected arrival units.

Changed retry

Move the SLA from 120 minutes to 180 minutes and explain the changed alert behavior without changing data.

Close this review now. Save retry under evidence/L04/retry_001/ and explain what changed.

R05 - Metadata, lineage and ownership

Open only after genuine P05 evidence exists.

A strong attempt should show

An OpenLineage-shaped metadata event plus owner, code version, input/output dataset identifiers and downstream blast-radius note.

Diagnostic questions

1. Did you declare the correct boundary/grain?
2. Does the evidence actually prove the claim?
3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit?
5. Did you keep secrets/PII out of evidence?

Common miss

Drawing arrows without owner/version; treating filename only as dataset identity; creating false all-to-all edges; confusing run with job.

Targeted repair

If lineage has arrows but no run/version/owner, it cannot support incident diagnosis or change blast-radius decisions.

Changed retry

Add a downstream daily sales asset and identify who must be notified for a breaking schema change.

Close this review now. Save retry under evidence/L05/retry_001/ and explain what changed.

R06 - Logs, metrics and tracing concepts

Open only after genuine P06 evidence exists.

A strong attempt should show

A signal contract naming required fields for logs, metrics and trace context without raw PII or secrets.

Diagnostic questions

1. Did you declare the correct boundary/grain? 2. Does the evidence actually prove the claim? 3. Could a green job still hide the failure? 4. Is owner/action/recovery explicit? 5. Did you keep secrets/PII out of evidence?

Common miss

Logging entire records; using metrics without labels; creating trace IDs but not logging them; putting secrets in error payloads.

Targeted repair

If a log says only error=true, add operation, dataset, run_id, error class and safe context. If it contains raw PII or credentials, redesign the evidence.

Changed retry

Add a second job in the same trace and state which fields should propagate across the boundary.

Close this review now. Save retry under evidence/L06/retry_001/ and explain what changed.

R07 - Pipeline SLIs, SLOs and alert design

Open only after genuine P07 evidence exists.

A strong attempt should show

At least two pipeline SLIs, explicit SLO windows, page/ticket thresholds, owner and recovery action.

Diagnostic questions

1. Did you declare the correct boundary/grain?
2. Does the evidence actually prove the claim?
3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit?
5. Did you keep secrets/PII out of evidence?

Common miss

Choosing 100% by instinct; alerting on raw CPU only; SLO without time window; pager without action.

Targeted repair

If your alert has no owner or action, it is only a notification. If your SLO is 100% by default, explain the operational/cost implication.

Changed retry

Change one failed run to a warning-only anomaly and decide whether the page still fires.

Close this review now. Save retry under evidence/L07/retry_001/ and explain what changed.

R08 - Incident triage and root-cause analysis

Open only after genuine P08 evidence exists.

A strong attempt should show

Timeline, impact, containment, evidence-backed cause, contributing factors, repair/reprocess scope, post-repair reconciliation and prevention.

Diagnostic questions

1. Did you declare the correct boundary/grain?
2. Does the evidence actually prove the claim?
3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit?
5. Did you keep secrets/PII out of evidence?

Common miss

Starting with blame; deleting evidence; reprocessing everything; closing when job turns green; writing "human error" as root cause.

Targeted repair

If your root cause is "bad data", ask what mechanism admitted it. If your fix is "rerun", ask why rerun is safe and how correctness is re-proved.

Changed retry

A second incident has correct data but missing telemetry: classify impact and preventive control.

Close this review now. Save retry under evidence/L08/retry_001/ and explain what changed.

R09 - Runbooks and operational handoff

Open only after genuine P09 evidence exists.

A strong attempt should show

Operator-ready runbook containing symptoms, owner, first checks, safe commands/actions, stop conditions, recovery, validation and escalation.

Diagnostic questions

1. Did you declare the correct boundary/grain? 2. Does the evidence actually prove the claim? 3. Could a green job still hide the failure? 4. Is owner/action/recovery explicit? 5. Did you keep secrets/PII out of evidence?

Common miss

Writing vague “restart pipeline”; no rollback/stop boundary; embedding secrets; no validation after action.

Targeted repair

If another engineer cannot decide whether a rerun is safe, add state/idempotence and reconciliation guidance.

Changed retry

Change the failure from missing row to expired credential and update only the relevant diagnostic/recovery branch.

Close this review now. Save retry under evidence/L09/retry_001/ and explain what changed.

R10 - Secrets, identity and access-control engineering

Open only after genuine P10 evidence exists.

A strong attempt should show

Least-privilege matrix, minimal workflow permissions, protected production environment, secret lifecycle and OIDC trust boundary.

Diagnostic questions

1. Did you declare the correct boundary/grain?
2. Does the evidence actually prove the claim?
3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit?
5. Did you keep secrets/PII out of evidence?

Common miss

Giving admin to make pipeline work; repo-wide long-lived secret; logging tokens; OIDC trust policy without branch/repo conditions.

Targeted repair

If a workflow needs broad write-all permissions, enumerate the exact API/action that needs each permission. If a secret appears in logs, treat it as compromised and rotate/revoke.

Changed retry

Move deployment from test to production and add required reviewer/environment restrictions.

Close this review now. Save retry under evidence/L10/retry_001/ and explain what changed.

R11 - Encryption and sensitive-data handling

Open only after genuine P11 evidence exists.

A strong attempt should show

Classification, encryption, access, masking, evidence-sanitization and retention rules plus one negative exposure test.

Diagnostic questions

1. Did you declare the correct boundary/grain? 2. Does the evidence actually prove the claim? 3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit? 5. Did you keep secrets/PII out of evidence?

Common miss

Assuming encryption grants authorization; copying raw PII into tickets; keeping secrets in logs; indefinite retention.

Targeted repair

If you call masking encryption, restate the threat model. If raw sensitive values are required for diagnosis, define secure access and retention rather than copying them to a report.

Changed retry

A regulator now requires 30-day incident evidence retention; update storage/access/expiry rules.

Close this review now. Save retry under evidence/L11/retry_001/ and explain what changed.

R12 - CI/CD for data projects and automated checks

Open only after genuine P12 evidence exists.

A strong attempt should show

A CI workflow/checklist where failing tests block promotion and PR jobs cannot access production credentials.

Diagnostic questions

1. Did you declare the correct boundary/grain?
2. Does the evidence actually prove the claim?
3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit?
5. Did you keep secrets/PII out of evidence?

Common miss

Deploying on every PR; storing prod secret at repo scope; ignoring failed tests; letting deployment start before approval.

Targeted repair

If deployment can run after failed tests, fix dependencies. If PR workflows can access production secrets, redesign the environment/trigger boundary.

Changed retry

Add a schema-contract change and define the required migration/reconciliation gate before deployment.

Close this review now. Save retry under evidence/L12/retry_001/ and explain what changed.

R13 - Environment promotion and rollback

Open only after genuine P13 evidence exists.

A strong attempt should show

Dev/Test/Prod promotion plan, configuration/identity boundaries, rollback-vs-roll-forward decision, smoke test and reconciliation.

Diagnostic questions

1. Did you declare the correct boundary/grain?
2. Does the evidence actually prove the claim?
3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit?
5. Did you keep secrets/PII out of evidence?

Common miss

Different code forks per environment; no post-deploy check; rollback after irreversible data change; manual prod edits not recorded.

Targeted repair

If rollback is your automatic answer, check schema/state compatibility and whether data already changed. If roll-forward, bound affected scope and prove repair.

Changed retry

The release changed only stateless validation thresholds: re-evaluate the decision.

Close this review now. Save retry under evidence/L13/retry_001/ and explain what changed.

R14 - Infrastructure-as-code concepts for data systems

Open only after genuine P14 evidence exists.

A strong attempt should show

Learning-only plan review explaining desired config, proposed actions, state sensitivity, locking, drift and safe review boundary.

Diagnostic questions

1. Did you declare the correct boundary/grain?
2. Does the evidence actually prove the claim?
3. Could a green job still hide the failure?
4. Is owner/action/recovery explicit?
5. Did you keep secrets/PII out of evidence?

Common miss

Treating plan as applied result; committing sensitive state; force-unlock without verifying owner; manual fix without updating code.

Targeted repair

If your plan file/state is committed casually, treat that as a security/operations defect. If drift is fixed manually without updating code, you have not restored declarative control.

Changed retry

A manual production change creates one drifted setting: describe plan/review/repair without applying real cloud resources.

Close this review now. Save retry under evidence/L14/retry_001/ and explain what changed.

R15 - Cost, performance and reliability trade-offs

Open only after genuine P15 evidence exists.

A strong attempt should show

ADR comparing options across correctness, latency, reliability, operator effort and cost, with non-negotiable security/correctness floors and review trigger.

Diagnostic questions

1. Did you declare the correct boundary/grain? 2. Does the evidence actually prove the claim? 3. Could a green job still hide the failure? 4. Is owner/action/recovery explicit? 5. Did you keep secrets/PII out of evidence?

Common miss

Optimizing only cloud bill; assuming maximum redundancy is always best; hiding correctness trade-offs; no review trigger.

Targeted repair

If the cheapest option wins automatically, quantify failure cost and SLO impact. If the most reliable option wins automatically, quantify marginal benefit and ongoing operational cost.

Changed retry

Traffic doubles while business freshness requirement stays unchanged: revisit the ADR.

Close this review now. Save retry under evidence/L15/retry_001/ and explain what changed.